EX : 7 Integrate a Vector Database with an LLM to build a Retrieval-Augmented Generation (RAG) system that answers questions based on external PDF documents.

Dataset: LinkedIn Profile or Research Paper PDFs.

AIM

To integrate a **Vector Database with a Large Language Model (LLM)** and build a **Retrieval-Augmented Generation (RAG)** system that answers user queries based on external PDF documents such as LinkedIn profiles or research papers.

CODE

```
# Step 1: Install required libraries (run once)
# pip install langchain faiss-cpu pypdf sentence-transformers openai

from langchain.document_loaders import PyPDFLoader
from langchain.text_splitter import CharacterTextSplitter
from langchain.embeddings import HuggingFaceEmbeddings
from langchain.vectorstores import FAISS
from langchain.llms import OpenAI
from langchain.chains import RetrievalQA

# Step 2: Load PDF
pdf_path = "sample.pdf"  # Replace with your PDF
loader = PyPDFLoader(pdf_path)
documents = loader.load()

# Step 3: Split text into chunks
text_splitter = CharacterTextSplitter(
    chunk_size=500,
    chunk_overlap=50
)
```

```
docs = text_splitter.split_documents(documents)

# Step 4: Convert text into embeddings
embeddings = HuggingFaceEmbeddings()

# Step 5: Store embeddings in FAISS vector database
vector_db = FAISS.from_documents(docs, embeddings)

# Step 6: Create retriever
retriever = vector_db.as_retriever()

# Step 7: Initialize LLM
llm = OpenAI(temperature=0)

# Step 8: Create RAG pipeline
qa_chain = RetrievalQA.from_chain_type(
    llm=llm,
    retriever=retriever
)

# Step 9: Ask question
query = "What are the key skills mentioned in the document?"
response = qa_chain.run(query)

# Step 10: Output result
print("Question:", query)
print("Answer:", response)
```

SAMPLE OUTPUT

Question: What are the key skills mentioned in the document?

Answer: The document highlights skills such as Python programming,

Machine Learning, Data Analysis, SQL, and Web Development.

RESULT

The system successfully:

- Extracted text from PDF documents
- Converted text into vector embeddings
- Stored embeddings in a FAISS vector database
- Retrieved relevant document chunks based on user queries
- Generated accurate answers using an LLM